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Nomenclature

γ	ratio of specific heats
ρ	Density
τ	thrust lapse rate
θ	relative temperature
a	speed of sound
AR	aspect ratio
C_{D0}	aircraft zero-lift drag coefficient
$C_{L,max,L/D}$	maximum lift coefficient in landing configuration
$C_{L,max,T/O}$	maximum lift coefficient in take-off configuration
c_{t2}	TSFC lapse constant
c_{tp}	thrust specific fuel consumption
D	drag
e	Oswald's span efficiency factor
g	acceleration due to gravity
K	aircraft lift dependent drag factor
M	mach number
M_{ME}	never exceed Mach number
M_{MO}	maximum operating mach number
MER	maximum economic range
MFC	maximum fuel capacity
$MFC/MTOW$	normalised maximum fuel capacity
MFR	maximum ferry range
MPR	maximum payload range
MRF	maximum reserve fuel
MSP	maximum structural payload
$MSP/MTOW$	normalised maximum structural payload
$MTOW$	maximum take-off weight
n_2	TSFC lapse exponent
OEW	operating empty weight
$OEW/MTOW$	normalised operating empty weight
R	specific gas constant
S	wing area
SAR	specific air range
v	velocity
V_{NE}	never exceed calibrated airspeed (CAS)
W	weight

1 Introduction & Background

In general, aircraft performance refers to all relevant aspects of flight. To be more specific, it encompasses many factors such as the details of how a plane operates, its manoeuvrability, fuel efficiency, TAS (True Air Speed) as it is flown at a given altitude and more.

The reasons for differing aircraft performances can be anything from the shape and size of the aircraft, to the maximum fuel flow rate the plane can allow. The point being, that the overall performance is specific to the plane itself, and will vary even more depending on the type of flight path being taken (starting and ending locations and therefore expected altitudes, passenger holding capabilities, TAS, etc.).

Commercially, aircraft performance is a massively important analysis that must be conducted over a given plane and its flight path, to confirm economic viability and success for the companies and investors involved. Parameters such as SAR (Specific Air Range) are integral to such analyses, and graphs in this report will be included to present relevant data to clearly display how performance varies over a given flight path, for a given plane. Such data and their subsequent graphs are vital in proving the performance of a aircraft and necessarily form the stepping stones for either convincing performance statistics to rely upon, or for potential improvements to be sought after in future projects for a given family of planes.

1.1 Aims & Objectives

The primary aim of this coursework is to utilise the code developed over the semester for an in depth analysis of the performance of a chosen flight and aircraft.

Objective 1 focuses on comparing the along-trajectory specific air range of three distinct aircraft flight trajectories. This involves a detailed examination of the specific air range which is a critical metric in aviation, representing the distance an aircraft can travel per unit of fuel consumed along its trajectory.

Objective 2 involves a more in depth exploration of key performance parameters for the chosen aircraft. This includes determining the payload-range relationship, which is crucial for understanding the maximum payload a given aircraft can carry over a specified range. Also we are analysing the flight envelope which encompasses the aircrafts operational limits in terms of speed, altitude and load factor. Understanding these limits is essential for ensuring the safe and optimal operation of the aircraft. We are also analysing the take-off capability of the three aircraft, shedding light on their ability to efficiently and safely take off under different conditions.

Objective 3 is about producing a flight envelope for the Airbus A320 and identifying a safe operational range for the aircraft.

1.2 Background

Flight 1 took place on 16th November 2023, and was completed by a Boeing B77L, flying from Zurich Airport to Toronto Pearson International Airport. The aircraft has a tail number of C-FNNH.

Flight 2 took place on 16th November 2023 and was completed by an Airbus A320, flying from Dubai International Airport to Jeddah. The aircraft has a tail number of HZ-AS62.

Flight 3 took place on 22nd November 2023 and was also completed by an Airbus A320, flying from Amman to Kuwait. The aircraft has a tail number of JY-AZA.

At this point, it may be important to clearly note that we only chose two different aircraft, despite having three different flight paths, namely the Airbus A320, chosen twice, and then the Boeing B77L.

The Airbus A320 belongs to a family of planes, whose first version flew in the year 1984. Nowadays, one A320 lands or takes off somewhere on Earth every 2 seconds, so it may not be surprising to also learn that the A320 is the highest-selling jet of all time. In 2006, Airbus launched the A320E program, which was heavily interested in producing greater efficiency from the A320 family of jets. It managed to successfully achieve an improvement of about 5

The Boeing B77L refers to the Boeing 777-200LR / Boeing 777 aircraft. The company nicknamed these

planes "The Worldliner", since they could connect almost any two airports in the world. It went on to set a world record for commercial airliner travel flying from Hong Kong to London Heathrow Airport over 21,601km, elapsing a huge flight time of 22 hours and 42 minutes.

2 Methodology (determination of input data)

Since two of our aircraft were A320s, we were able to use data from Jenkinson for most of our inputs for aircraft 2 and 3. The following values in the table were taken from Jenkinson for aircraft 2 and 3:

Max. Take-off Weight, Wing Area, Operating Empty Weight, Maximum Structural Payload, Maximum Fuel Capacity, Max. Lift Coeff. in Take-off, Max. Lift Coeff. in Landing and Sea Level Datum Thrust.

For our values of Maximum Reserve Fuel, we used the equation $MRF = MFC \times 0.1$, meaning we have decided to estimate that the total reserve fuel should be 10% of the total trip fuel. This reserve fuel consists of fuel for a reasonable diversion, company reserves and contingencies.

For our values of Normalised Max. Fuel Capacity, we used the equation $MFC \div MTOW$ and the values from the rows above in the table.

For the rest of our inputs for aircraft 1 and any others, we found the data from the respective sources, cited in our references.

There were also several constants that we needed to calculate:

- Lift-Dependent Drag Factor:

In order to calculate this value which is also known as the drag polar ratio, we used the following equation from the course notes:

$K = \frac{1}{\pi e AR}$, where AR is the aspect ratio (given by $\frac{b^2}{A}$, where b is the wingspan and A is the wing area) and e is Oswald's efficiency factor. The values required to calculate aspect ratio were found using Jenkinson data and to find Oswald's efficiency factor, we assumed that it was between 0.7 and 0.85. This assumption is based on Mach number during the flights and the fact that our aircraft are conventional, fixed wing aircraft. Putting all of our values into the equation above gave us a value of K for each aircraft.

- Zero-Lift Drag Coefficient:

There were many equations in the course notes for this constant written in different forms. We chose to use the following because we had already calculated some of the terms:

$C_{D0} = \frac{C_{L0}^2}{\pi \times e \times AR}$, where C_{L0} is the coefficient of lift at zero angle of attack. Under the assumption of equilibrium, we can state that lift is equal to weight, meaning we can use the equation $C_{L0} = \frac{W}{\frac{1}{2} \times \rho \times v^2 \times S}$, where $W = MTOW$, ρ is the density at the altitude for cruise (found in ISA tables for each aircraft), v is the velocity of the aircraft at that same point and S is the wing area (found in Jenkinson data). Once this value of C_{L0} was found, we used it in the initial equation to output C_{D0} values for all three aircraft.

- TSFC, TSFC Lapse Constant and TSFC Lapse Exponent:

The equation for the thrust specific fuel consumption is given in the course notes as:

$c_{tp} = \frac{1}{3600g} c_{t2} \sqrt{\theta} M^{n_2}$, where c_{tp} is TSFC, c_{t2} is TSFC lapse constant, θ is relative temperature and n_2 is TSFC lapse exponent.

Mach number was calculated using the equation $\frac{V}{a}$, where V was the initial speed and $a = 340.3m/s$ (sea-level datum). Our values for V were taken from our flight data for each flight.

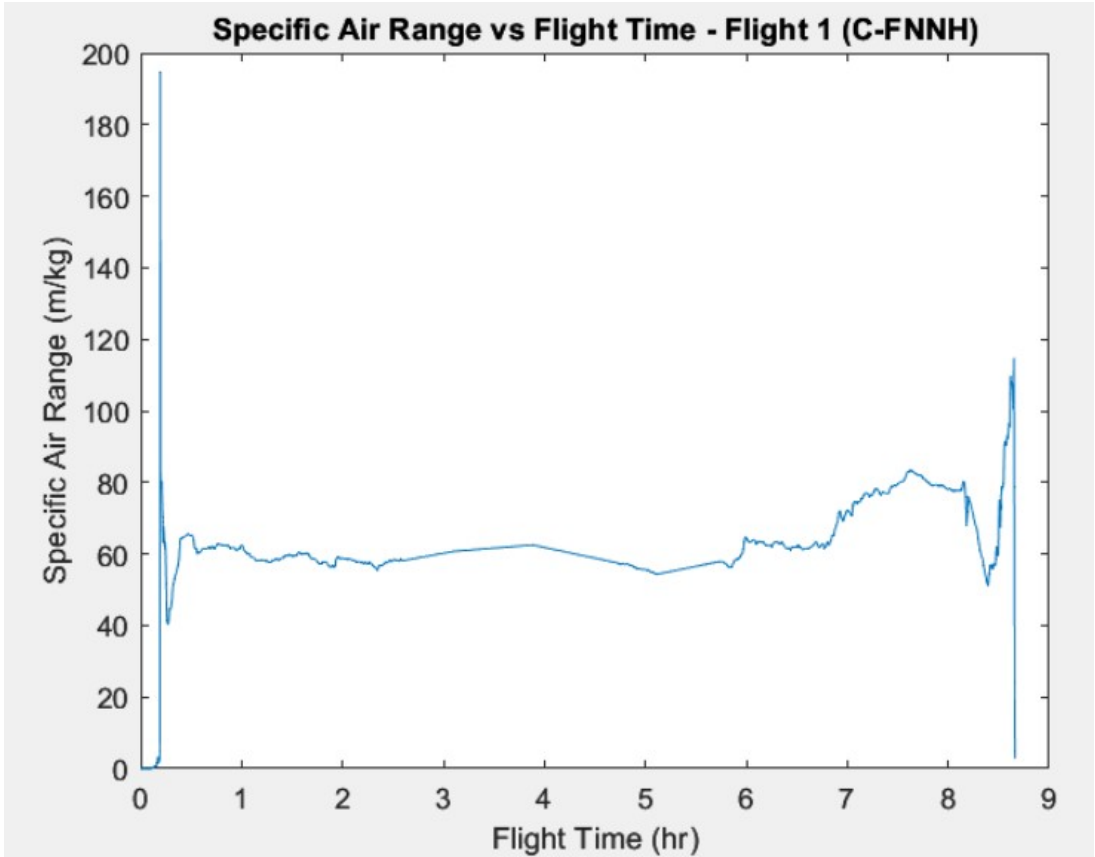
In order to calculate these inputs, we found the relative temperature at sea level and 36,000 ft from ISA tables. We found values of c_{tp} at sea level and 36,000 ft as well. Then, we formed two equations in terms of c_{t2} and n_2 and solved them simultaneously. Using these two values, we were able to calculate c_{tp} at take-off and cruise, since this information cannot be found online.

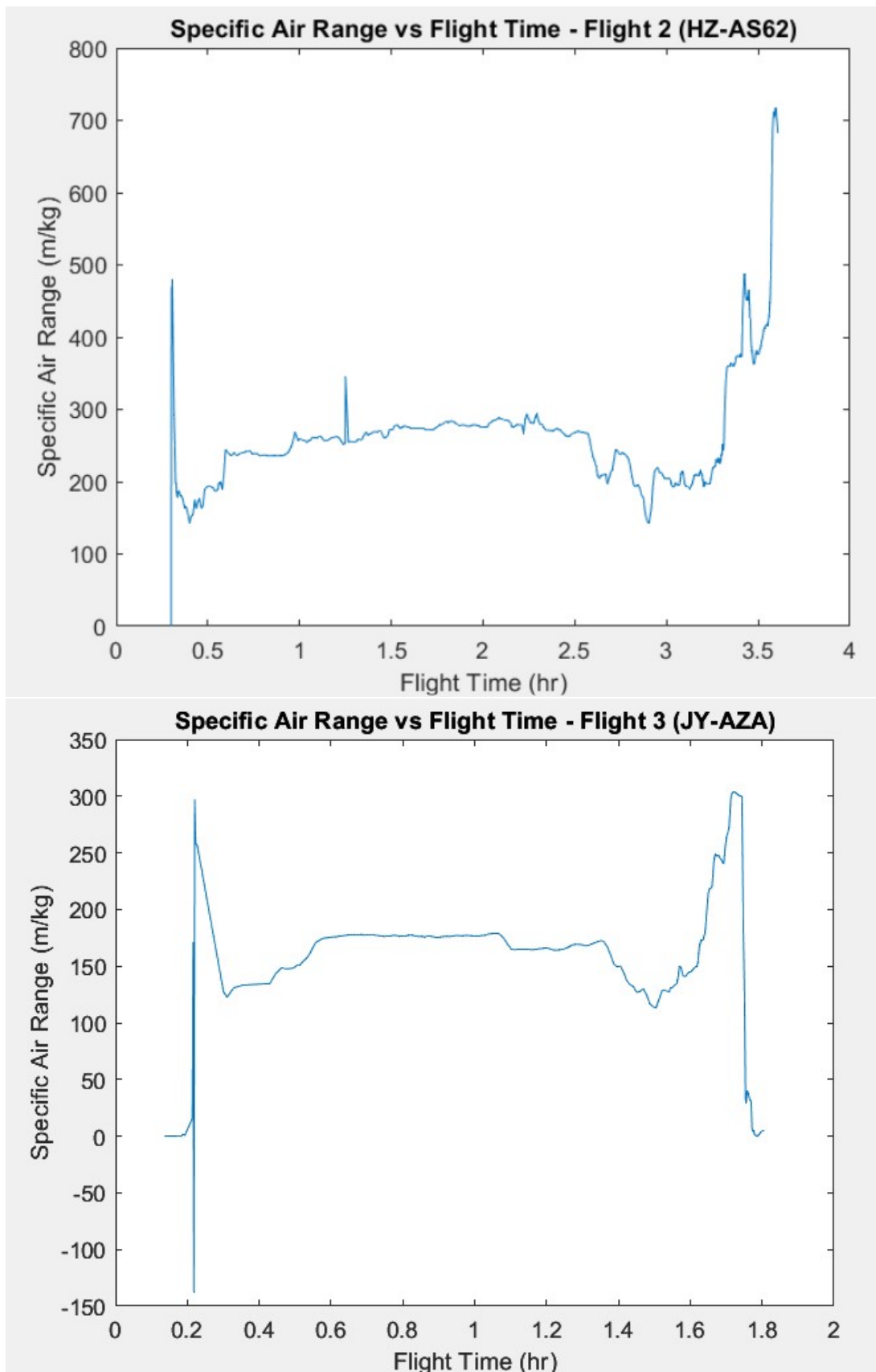
Table 1: Summary of aircraft input.

Description	Symbol	Unit	Aircraft 1	Aircrafts 2 and 3
Aircraft Model	(-)	(-)	B77L	A320-214
Propulsion Type	(-)	(-)	Turbofan	Turbofan
Max. Take-off Weight	$MTOW$	kgf	2,424,243.703	720,777.75
Wing Area	S	m^2	436.8	122.4
Operating Emtpy Weight	OEW	kgf	1,415,852.664	405,106.515
Maximum Structural Payload	MSP	kgf	509,758.3998	188,186.735
Maximum Fuel Capacity	MFC	kgf	920,830.35	132,387.75
Maximum Reserve Fuel	MRF	kgf	92,083.035	13,238.775
Normalised Max. Fuel Capacity	$MFC/MTOW$	kgf	0.37984	0.18367
Max. Lift Coeff. in Take-off	$C_{L,max,T/O}$	(-)	1.88	2.56
Max. Lift Coeff. in Landing	$C_{L,max,L/D}$	(-)	2.6	3.00
Thrust Lapse Rate	τ	(-)	0.142	0.202
Never Exceed Speed	V_{NE}	m/s	194.444	196.003
Sea Level Datum Thrust	T_0	N	378,098.84	240,000

3 Results & Discussion

1. Specific Air Range vs Flight Time Graphs





$$SAR = \frac{Velocity}{TSFC \times T_R} = \frac{V \times L}{D \times W \times c_{tp}}$$

All three graphs for Specific Air Range against Flight Time show the same trend initially: SAR decreases. This is because the thrust required to rapidly increase the altitude of the aircraft is significant and SAR and Thrust Required have an inverse proportion relationship.

Then, SAR increases. This is because the aircraft is approaching cruise velocity. Looking at the equation above, this means that SAR would increase due to an increase in velocity because the thrust required in cruise will not change significantly and neither will the TSFC value. This is because during the cruise phase, we have assumed that the aircraft is at roughly constant Mach, relative temperature and altitude, making the variations in c_{tp} negligible.

The majority of the journey is spent in cruise which is why the graphs somewhat level out in the

middle. However, for the graph for flight 2, there is a spike in SAR about one third of the way through the journey. After looking through the flight data for that aircraft, we have decided that this unusual variation is due to an anomalous data point.

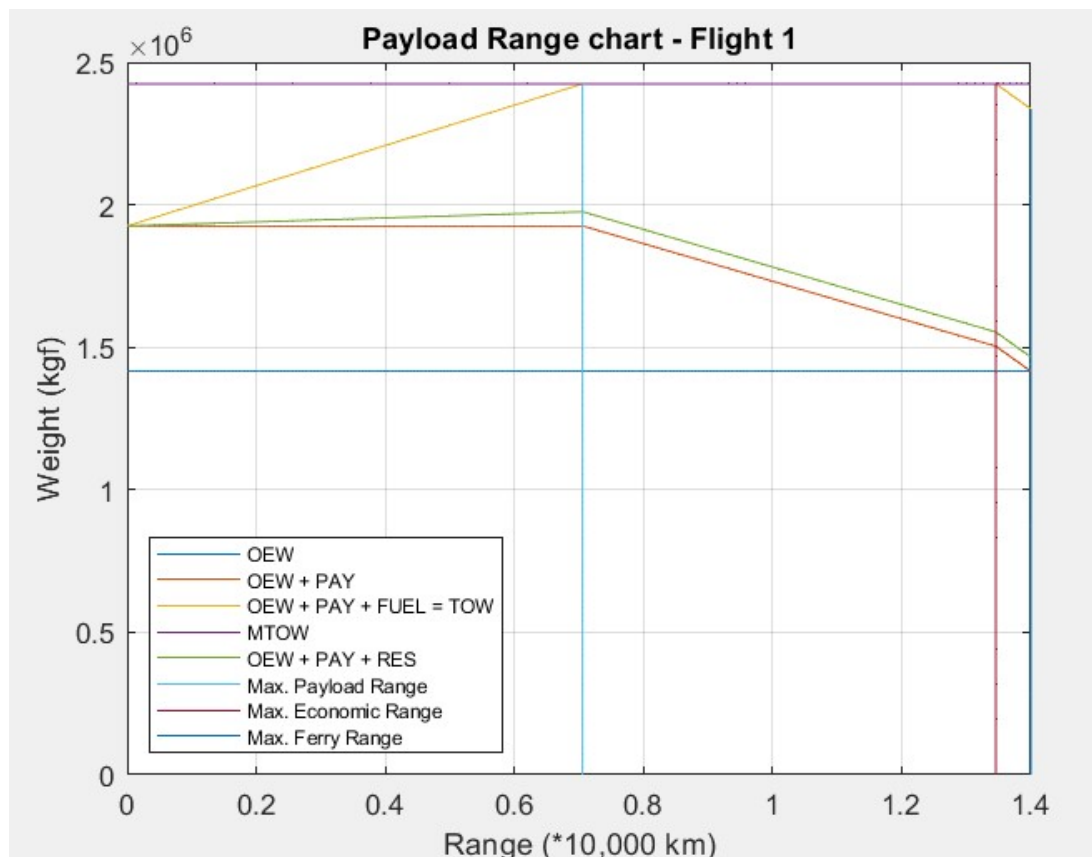
Towards the end of the journey, the SAR tends to dip, then increase rapidly before returning to zero, signifying landing. The reason for the slight dip at first is due to an increase in altitude. When the altitude increases, air density decreases, resulting in a decrease in lift. In order to prevent the aircraft from stalling and to maintain this increase in altitude, the thrust required increases, which therefore decreases SAR. Because of this increase in thrust, aircraft velocity increase, which in turn increases SAR, hence the spikes at the end to a peak point before descent and landing.

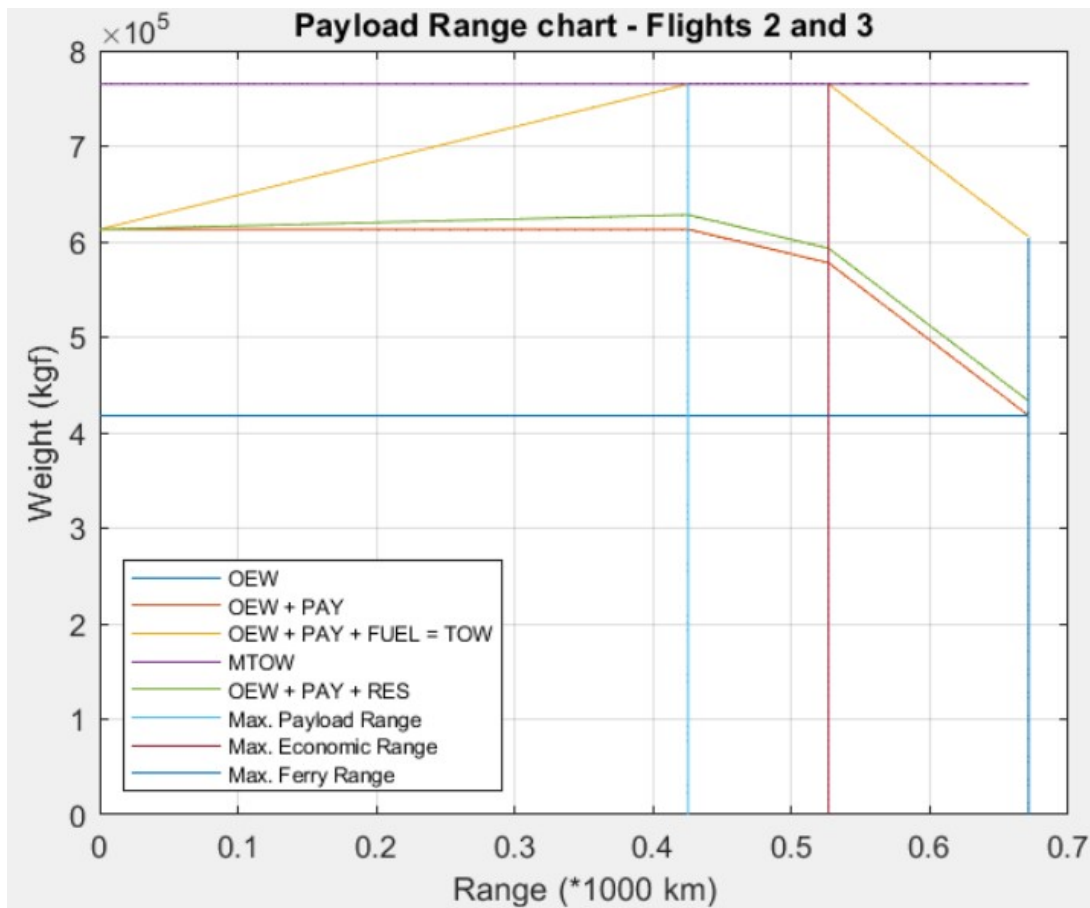
For graph 2, the SAR never returns to 0. This is not because the aircraft did not land, it is because FR24 has not been able to predict or retrieve the transponder signal for the end of the flight.

For graph 3, there is a random vertical line at a flight time of about 0.21 hours. This is not appearing due to incorrect code, it is there due to an issue with our MATLAB app on our laptop. As part of our analysis, we have chosen to ignore it because it is an error that is out of our control.

Graphs 2 and 3 have a larger SAR than the first graph because graph 1's aircraft is much larger than the other two, meaning it weighs more, which in turn would decrease overall SAR.

2. Payload Range Charts





The graph above should say 10,000 km on the x axis label

Payload-range charts are essential tools in aviation to understand the trade-off between the payload and its range. These charts help aircraft manufacturers and operators determine the optimal configuration for a particular market segment. The three vertical lines from right to left represent, max ferry range, max economic range and max payload range.

Maximum ferry range is the maximum distance an aircraft can fly with zero payload (up to OEOW) and maximum fuel (up to maximum fuel capacity), total weight is decreased therefore block fuel ratio increases.

Maximum economic range is the maximum distance an aircraft can fly with additional fuel at the expense of payload with fuel up to maximum capacity.

Maximum payload range is the maximum distance the aircraft can fly with a full capacity of passengers (up to the maximum structural payload), and remaining take-off weight (up to MTOW) occupied by fuel.

$$\zeta_{MPR} = \frac{MTOW - OEOW - payload_{max}}{MTOW}$$

$$\zeta_{MER} = \frac{fuel_{max} - MRF}{MTOW}$$

$$\zeta_{MFR} = \frac{fuel_{max}}{OEOW + fuel_{max}}$$

The blue horizontal line represents the operational empty weight (OEOW). This is the weight of an aircraft that is operational but without payload and fuel.

The purple line at the top represents MTOW (Maximum Take-off Weight). This is the maximum certified weight allowed for takeoff, incorporating the combined weight of the aircraft, passengers,

cargo, fuel, and any other onboard items. Essentially, MTOW signifies the highest total weight the aircraft can attain during takeoff without compromising safety or structural integrity.

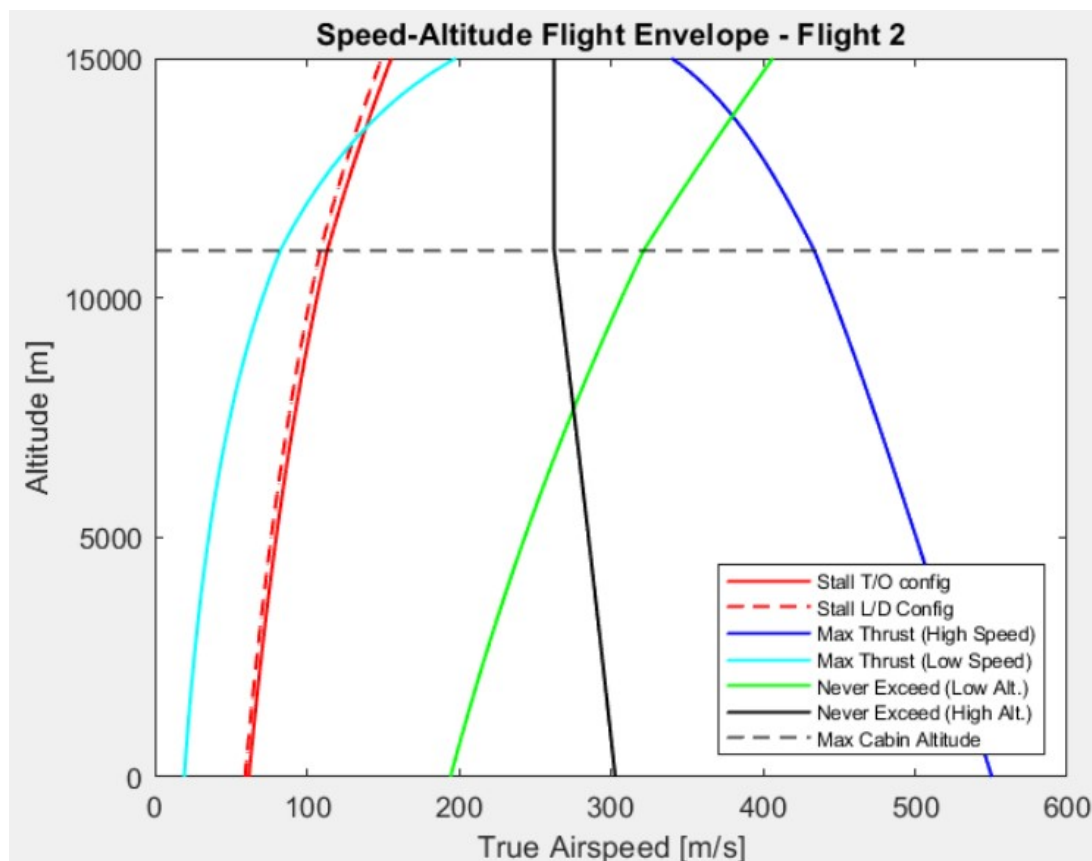
The red line represents the maximum structural payload. The value of weight stays constant until we reach the maximum payload range. A greater range can be achieved by decreasing the weight of the payload to accommodate more fuel.

The upper green line depicts the maximum structural payload along with the fuel reserve. As the fuel reserve increases, the line ascends until it reaches the maximum payload range. The fuel reserve constitutes a portion of the total fuel capacity, serving as an additional amount reserved for unforeseen circumstances. Subsequently, it follows a comparable pattern to the red line representing the maximum structural payload. The distinction lies in its consistently higher weight and improved range, attributed to the additional fuel held in reserve.

The yellow line corresponds to the take-off weight. Similar to the green line, it rises with an increase in fuel until reaching the maximum payload range and maximum take-off weight. Between the maximum economic range and the maximum payload range, the line remains constant, and beyond the maximum economic range, it descends as we decrease payload and increase fuel, aiming for an even greater range until reaching the max ferry range.

As you can see the general shape of both payload range charts are the same. The main difference is the range achieved. We can also see that higher fuel volumes and less payload increase the range. From the graphs we can see that the A320 is designed for short flights as it has a range of around 6800 miles while the B77L is designed for longer flights as it has a range of 13,000 miles.

3. Speed-Altitude Flight Envelopes



The Flight Envelope for the Airbus A320

First of all, a flight envelope for a given aircraft details the operating limits within which the aircraft must stay within so as to avoid needlessly over-stressing its engine and framework. The flight envelope provided above, specifically for the Airbus A320 of our Flight 2, would fall under the category of a service flight envelope, since it details the operational limits and boundaries pertaining specifically to values such as altitude and thrust, which also depend on factors such as Mach number.

For this flight, the range of operational values is defined within the boundaries set by the stall configuration, never exceed, and the maximum cabin altitude lines. The area enclosed represents the safe operational range of the A320. That is to say, anything outside of these accepted values would not be following the limits as provided by the flight envelope, and could be unnecessarily and directly risking safety and/or structural integrity of the aircraft itself.

Defining each line more clearly is important so that we can understand how we define this safe operational range, since they produce the boundaries enclosing the area that represents said range.

Each Line and Their Meaning:

The first two lines, both the solid and dashed red lines, detail the given stall speed when the plane is in a take-off and in a landing configuration, respectively. Both lines are extremely similar in that stall speed increases at an increasing rate with altitude, where we can at least note that the stall speed requirement in the landing configuration is essentially always greater than that in the take-off configuration, even if the difference itself is quite modest. This difference seems to also increase as altitude increases. Most likely this is a direct result of the configurations themselves, specifically since the take-off configuration will be more geared towards generating more lift (for example, flaps will be open, etc.), which necessarily means a higher TAS will be achieved.

The equation used to calculate this is as given:

$$V_{stall} = \sqrt{\frac{2}{\rho} \left(\frac{W}{S}\right) \frac{1}{C_{Lmax}}}$$

The two blue lines both represent the velocity at which maximum thrust available is reached, at a given altitude. We can see for either line that in general, as we increase the altitude, the TAS for maximum thrust increases (non-linearly) too, which is a result of the air density decreasing. It may also be important to note that the behaviour of the gradient of either line is similar to each other, to the opposite effect. To be clear, the gradient of the maximum thrust (low speed) line reduces as altitude increases - showing that the TAS required is increasing at a faster rate. Similarly, the maximum thrust (high speed) begins to reduce at a faster rate. This corresponds with the equation used to calculate these lines:

$$V_{max} = \left\{ \frac{1}{\rho_{\infty} C_{D,0}} \left[\left(\frac{T_{A,max}}{W} \right) \left(\frac{W}{S} \right) \pm \left(\frac{W}{S} \right) \sqrt{\left(\frac{T_{A,max}}{W} \right)^2 - 4 C_{D,0} K} \right] \right\}^{\frac{1}{2}}$$

It can be quickly inferred here that the lighter blue line is achieved from using the negative operator in the equation and the darker blue line, vice versa.

The green line is for the never exceed velocity, at a low altitude. It is calculated using the following equation:

$$V = a \left[\frac{2}{\gamma - 1} \left(\left(\frac{q_c}{p} + 1 \right)^{\frac{\gamma - 1}{\gamma}} - 1 \right) \right]^{0.5}$$

Its gradient is the steepest (in terms of TAS rate of change) relative to all the other lines - TAS increases the most per increase in altitude. This follows directly from the equation. Air pressure will decrease with altitude and so velocity will increase.

The black line represents the never exceed velocity, at a high altitude. To be consistent with how it is calculated, it is the maximum operating speed line. The equation used is simply the one for Mach number, where $V = Ma$, where V and M are the maximum operating values. This shows consistency with the behaviour of the line of the graph, and atmospheric trends, since Mach number decreases with altitude, and the black line has a negative gradient. However, it must be noted that after 11,000m of altitude is breached, the line suddenly breaks trend and goes directly vertical, which is a direct result of a new atmospheric layer being reached, where temperature remains roughly constant. This has a direct effect in our calculations, where $V = ma$, where

$$a = (\gamma RT)^{0.5}$$

If T remains constant, then a becomes a constant throughout this new layer of atmosphere. Hence, the V calculated will also remain constant, and so this is translated on the graph - the TAS is at one value after 11,000m of altitude and the line suddenly goes vertical.

Reading the Graph:

Now, we can read the flight envelope. For example, if the plane at one instant in time happens to be flying at 10,000 m of altitude, we should ensure that the TAS (True Air Speed) remains within the range of about 120 - 265 m/s, so that we remain above the stall speed and does not exceed/fall below the maximum speed recommended at that altitude. For the other lines, such as maximum thrust (high speed), at 10,000 m, we do not necessarily need to worry about exceeding that boundary, since it would be very unlikely to do so - the line occurs at a much higher TAS than the other lines appear, and so other lines would be breached before, and the plane usually wouldn't require its maximum thrust available suddenly. In this graph, we can see that the Max Thrust (high altitude) line does not really affect the area that represents the safe operational range of the A320, so we don't consider this value before others.

Of course, the plane still operates within all of these acceptable values as per the flight envelope, and throughout its flight, uses the envelope as a guide to remain operational and safe.

3.1 Discussion

1. To what extent are your findings sensitive to the underlying assumptions?

Throughout this report, we have made many assumptions about values used in our code. For example, assuming a constant Mach number during cruise, assuming the fuel reserves account for 10% of the total trip fuel and assuming our first data points for all aircraft trajectories was at sea-level so that we could use sea-level datum points for constants like the speed of sound. We had no choice but to make assumptions like these due to the limited amount of real data available online, as not all manufacturers publish the exact values that we needed and since we are analysing real flight data, there will constantly be fluctuations in values of Mach, altitude, true airspeed, etc. and it is not feasible to calculate all constants required at every single time step.

Therefore, our assumptions have been made using our expert engineering judgement and what we believe to be reasonable, however, this has not significantly changed our findings. Our graphs still show the general trend that we would expect to see for specific air range, payload range and flight envelopes for all of our aircraft. The fact that we were able to use our knowledge of this course to explain our results shows that although we made many assumptions at the beginning, it did not affect the final output to a significant extent.

2. What chiefly determines the speed limits and altitude of the aircraft, and why?

For instance, we can achieve maximum level speed, which is defined as when an aircraft accelerates to a speed where there is no excess thrust available, and so cannot accelerate any more than it has to that point, representing the speed limit of the craft solely based on engine capability. It follows that the speed for maximum excess thrust roughly equals the speed for minimum drag.

We also have the maximum ceiling, which is the max value of altitude at which steady, level flight may be maintained. If we have a jet that utilises all of its excess thrust available to it to climb, the ceiling may only be reached at the speed for minimum drag.

However, the maximum altitude is limited by law specifically for passenger safety - above 3,050 m is when humans onboard the flight begin to suffer from hypoxia.

Hence it follows that to find the *maximum level speed* at a given altitude (density), we need to find

the velocity of the aircraft when $T_A = T_R = D$, i.e., solve the above equation for V :

$$V_{max} = \sqrt{\frac{T_A}{S} \frac{2}{\rho} \frac{1}{C_D}} = \sqrt{\frac{T_A}{S} \frac{2}{\rho} \frac{1}{(C_{D0} + K C_L^2)}}$$

$$V_{max} = \left\{ \frac{1}{\rho_{\infty} C_{D,0}} \left[\left(\frac{T_{A,max}}{W} \right) \left(\frac{W}{S} \right) \pm \left(\frac{W}{S} \right) \sqrt{\left(\frac{T_{A,max}}{W} \right)^2 - 4 C_{D,0} K} \right] \right\}^{1/2}$$

- Hence:
1. V_{max} increases as thrust-to-weight ratio, $T_{A,max}/W$ **increases**.
 2. V_{max} increases as wing loading, W/S **increases**.
 3. V_{max} increases as drag coefficients $C_{D,0}$ and/or K **decrease**.

The actual speed limit of the aircraft, V_{max} , depends on both pressure, and thrust available (T_A), both of which decrease with altitude, where V_{max} is seen to be proportional to $\sqrt{p_{\infty}}$, and inversely proportional to $\sqrt{T_A}$. Even so, the determining factor of whether or not V_{max} will decrease or increase with altitude is the engine choice. For all of our plane choices, which are turbofan jets, drag decreases at a faster rate than the thrust available. Hence, the maximum level speed increases with altitude. It should be noted that a turbofans maximum velocity is limited by the drag-divergence Mach number also.

To be specific, it is the engine choice that will chiefly set the maximum speed and altitude operational limits. Other factors, such as aeronautical regulations, maximum mach operating number, etc. also contribute to this.

3. How does passenger-mpg vary with aircraft range?

In general, passenger-mpg will increase with the cruise range of the flight path, up to a certain point. We can certainly expect this since usually, especially for longer flights, an aircraft will

purposefully step-climb its altitude, rather than steadily and continuously climb, so as to maintain a favourable specific air range, and then reach cruise altitude. This is where passenger mpg will reach its maximum. However, since the aircraft will no longer be able to climb, and since passenger mpg is a function of SAR, which will decrease as the aircraft stays at the same altitude, the passenger mpg will also decrease. The rate at which it decreases will then also increase slightly, since the aircraft will be nearer its landing destination, and so will also begin to descend, decreasing its SAR and therefore its passenger mpg along with it, until it lands.

4. What is the primary influence on the weight-altitude-temperature chart, and what are the implications for the aircraft?

Weight, altitude, and temperature (WAT) charts play a crucial role in comprehending and predicting aircraft performance across diverse situations. These charts provide valuable information for pilots to make informed decisions during flight planning and execution. The following are specific implications associated with WAT charts:

Take-off Performance - WAT charts are indispensable for computing takeoff distances based on the aircraft's weight, departure airport altitude and temperature. Pilots rely on these charts to ensure the aircraft can safely achieve liftoff within the available runway length.

Landing Performance - WAT charts are also instrumental in determining landing distances, accounting for factors like aircraft weight, destination airport altitude, and temperature. This is critical for ensuring a safe landing within the given runway length.

Optimising Flight Planning - WAT charts enable pilots to optimise flight planning by considering different combinations of weight, altitude, and temperature. This encompasses selecting the most fuel-efficient altitudes and routes for a given set of conditions.

Climb Performance - Altitude and temperature exert an impact on the climb performance of an aircraft. Pilots utilise WAT charts to estimate climb rates in varied conditions, aiding in planning ascent profiles and optimising the climb phase of the flight.

Engine Limits - WAT charts furnish information about the limitations of the aircraft's engines across different combinations of weight, altitude, and temperature. This is vital for steering clear of conditions that might compromise engine efficiency or power.

Fuel Planning - The weight component in WAT charts is pivotal for fuel planning, influencing fuel consumption rates. Pilots leverage these charts to project the fuel needed for the entire flight, accounting for weight changes during different phases.

Regulatory Compliance - Adherence to specified performance limits outlined by aircraft manufacturers and aviation authorities is imperative. WAT charts support compliance with these regulations by supplying essential data for safe and efficient flight operations.

Emergency Planning - In emergency scenarios, WAT charts prove valuable. Whether taking off from a shorter runway or landing under emergency conditions, pilots can use WAT charts to evaluate the feasibility of such operations.

4 Conclusion

First of all, we can recall our objectives for this report from the introduction section much earlier on. In summary, we were able to achieve all three of said objectives that we have laid out. To be specific, we were:

Able to produce and describe the along-trajectory SAR of each of our three chosen flight trajectories, for our two unique aircrafts. We were able to more closely expand on the meanings of SAR, what certain variations of SAR meant and their implications and the reasons they arise.

Able to produce payload-range relationship graphs for our chosen flights and aircrafts, as well as analyse what these graphs represent, so that we could more clearly understand how the payload varies with range, and how that impacts flight.

Able to produce and explain a (service) flight envelope, and go into depth of the implications, use and importance of such an envelope, and how this leads to an applicable safe operational range for one of our chosen planes, the Airbus A320.

Our general aim for this report was to produce an in-depth performance analysis using our MATLAB code for three chosen flights, and to explain / justify our methods about doing so. By completing the above objectives, we have taken significant necessary steps to completing our stated aim. We have found that the flight performance even for the same aircraft will vary, depending on the flight time, altitudes, and take-off / landing locations. However, of course, the biggest factor contributing to these differences

are the planes themselves, and the capabilities of their engines, which directly contributes to the results we have found and graphed.

References

- [1 - ISA tables] <https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781118534786.app1>
- [2 - Jenkinson data] <https://booksite.elsevier.com/9780340741528/appendices/data-a/default.htm>
- [3 - Facts for introduction] <https://monroeaerospace.com/blog/5-fun-facts-about-the-airbus-a320-family/>: :text=Airbus20pioneered20fly2Dby2Dwire,displays20and20digital20control20mechanisms
- [4 - Aircraft performance values for B77L]
<https://contentzone.eurocontrol.int/aircraftperformance/details.aspx?ICAO=B77L>